

F50 — Investigation #1 (FK matrix element factorization), driven to closure: the muon and CKM go OPPOSITE ways. CKM = two-point (N_c^2 amplitude = λ ; Direction B). Muon = single-object T190 (Composite v0.5 additive RETIRES under Reading (a); coupled gradings + T190 fits 40× better). Refines ‘one computation decides both.’

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2026-06-06 next-session, 2026-06-06 14:05 EDT

F50 — Investigation #1: FK Factorization, Both Sectors

0. The result, up front

Driving Investigation #1 to closure, the two sectors it was supposed to decide jointly go opposite ways — which itself refines Elie’s “one computation decides both” synthesis:

- CKM: two-point (irreducible). The mixing amplitude carries N_c^2 from two independent endpoint color traces. $\sin \theta_C = N_c^2 / (2^{\{N_c\} \cdot n_C}) = 9/40 = \lambda$ exactly. Single-trace (N_c^1) gives 0.075 — rejected. Direction B confirmed (FORWARD-derived).
- Muon: single-object. The mass ratio is the single-object T190 form $(24/\pi^2)^{\{C_2\}}$, not the additive Composite v0.5. Three converging lines (below). Composite v0.5’s additive form retires to T190 under Reading (a).

Same *kind* of question (does the FK matrix element factor through a single Bergman autocorrelation, or require an irreducible two-point structure?), opposite *answers*, in different *spaces* (CKM: color; muon: H^2 grading). Not one matrix element — two analogous factorization questions.

1. The factorization dichotomy (setup)

The FK matrix element $\langle V_\lambda | O | V_\mu \rangle$ on $H^2(D_{IV}^5)$: - Single-trace (Toeplitz / autocorrelation): $O = T_\varphi$ (multiply by symbol φ , project back to H^2). Then $\langle V_\lambda | T_\varphi | V_\mu \rangle = \int \bar{e}_\lambda \varphi e_\mu d\mu_B$ — one bulk integral, mediated by the Bergman reproducing kernel (the resolution of identity). One color trace for a color-neutral symbol $\rightarrow N_c^1$. - Two-point (irreducible): O is a genuine two-point object $O(z_1, z_2)$ that does *not* collapse to a single Bergman-mediated integral — two distinct endpoint contributions. Two independent color sums $\rightarrow N_c^2$.

The question for each sector: which holds?

2. CKM $\rightarrow$ two-point (Direction B, FORWARD-derived)

The Cabibbo amplitude mixes gen-1 and gen-2 down-type quarks (distinct generation K-types), each a color triplet. Test against the observed value:

structure	color power	$\sin \theta_C$ predicted	vs observed $\lambda = 0.2250$
single-trace (Toeplitz, N_c^1)	N_c^1	$N_c / (2^{\{N_c\} \cdot n_C})$ $= 3/40 = 0.075$	rejected (67% off)
two-point (irreducible, N_c^2)	N_c^2	$N_c^2 / (2^{\{N_c\} \cdot n_C})$ $= 9/40 = 0.2250$	exact (0.000%)

So the CKM mixing matrix element does not factor through a single color-neutral Toeplitz autocorrelation — it requires the irreducible two-point structure with two independent endpoint color traces (gen-1 color $\times$ gen-2 color). Direction B confirmed, FORWARD-derived from the observed value. The two quark endpoints are distinct boundary points whose color indices are summed independently — this is the substrate-mechanism content of the N_c^2 in the amplitude (F39's fork, resolved to B). Color here is *physical* (quarks).

3. Muon $\rightarrow$ single-object T190 (Composite v0.5 additive RETIRES)

Three converging lines drive the muon to a single-object form, against the additive Composite v0.5:

(i) Reading (a) retires the additive split (F44). $m_\mu/m_e = 1575/8 + 81/8$ as “bulk + boundary” double-counts two realizations of one H^2 object. Legitimate additivity needs the two terms to be distinct *orthogonal* structures (Investigation #2).

(ii) The gradings are (structurally) coupled, not orthogonal. The additive split needs the K-type grading (compact H_B , diagonal in K-types) and the restriction grading (Casey #14 $SO(5,2) \rightarrow SO(3,1)$, $P_{\text{restriction}}$) to commute. But $P_{\text{restriction}}$ projects

via a *noncompact* chain, and $SO(3,1) \not\cong K = SO(5) \times SO(2)$, so $P_{\text{restriction}}$ *mixes* K-types — hence $[H_B, P_{\text{restriction}}] \neq 0$ generically $\rightarrow$ coupled, not orthogonal $\rightarrow$ the additive “+” is illegitimate. (Structural argument; the explicit commutator is Investigation #2, handed to Elie for numerical confirmation — Sec. 5.)

(iii) T190 fits 40 $\times$ better, and is a single object. $T190 = (24/\pi^2)^{\{C_2\}} = 206.761$ (0.003% vs observed 206.768) vs Composite v0.5 additive = 207 (0.112%). T190 is a *single* transcendental Weyl-structure form ($24 = N_c \cdot |W(B_2)|$, the Weyl-orbit reading) — no additive split, no bulk+boundary, Reading-(a)-safe. (Precision corroborates but does not force — Cal #35; the load-bearing reasons are (i)+(ii). I flag that 0.003% sits below the Two-Tier structural floor, so I weight it as corroboration, not proof.)

Verdict: the muon mass operator factors as a single H^2 object, not an additive two-structure. Composite v0.5’s additive form retires to T190 — exactly the fallback F45 anticipated, now selected. Color does not enter (the muon is a singlet, F48); the “ $81/8 = N_c^4$ ” was never a color factor, and under the single-object reading it is not an additive term at all.

4. Why the two sectors diverge (the refinement of Elie’s synthesis)

Elie’s synthesis (“one FK computation decides both muon and CKM”) is the right *instinct* — both are factorization questions — but they are not one matrix element and they do not co-resolve:

- CKM factorization lives in color space (do the two quark endpoints’ color sums collapse to one trace?). Answer: no $\rightarrow$ two-point $\rightarrow N_c^2$. Color is physical.
- Muon factorization lives in H^2 grading space (do the K-type and restriction gradings commute?). Answer: (structurally) no $\rightarrow$ coupled $\rightarrow$ single object $\rightarrow$ T190. Color is absent (lepton).

Same shape (factor or not), opposite outcome (CKM irreducibly two-point; muon a single object), different space (color vs grading). The honest correction: two analogous factorization questions, separately resolved, not one computation. They *rhyme*; they don’t coincide.

5. Closure status + handoff

- CKM (Wall 5): CLOSED at Direction B (two-point, N_c^2 amplitude = λ ; single-trace rejected). FORWARD-derived. $\rightarrow$ @Grace R-6/d² reconciliation: d² = second-moment is the two-point/B branch (your Bergman=Born finding); R-6 promotes-or-retires on this. $\rightarrow$ @Keeper K234.
- Muon (Walls 2/3): T190 strongly indicated (Reading (a) + coupled gradings + 40 $\times$ better fit); Composite v0.5 additive RETIRED. Final confirmation = Investigation #2’s explicit $[H_B, P_{\text{restriction}}]$ commutator (I’ll set it up; @Elie numerical on $V_{(1/2,1/2)}$, $V_{(3/2,1/2)}$, $V_{(5/2,1/2)}$). If

[H_B,P_restriction] $\neq 0$ (expected), muon = T190 confirmed. $\rightarrow$ @Keeper K232 (muon re-derivation $\rightarrow$ T190).

- Refinement filed: “one computation decides both” $\rightarrow$ two analogous factorization questions, opposite answers, different spaces.

6. Honest tiering

- CKM Direction B: FORWARD-derived (observed value forces N_c^2 two-point; single-trace rejected at 0.075). Strong.
- Muon $\rightarrow$ T190: strongly indicated by three converging lines; the structural coupling argument (ii) is not yet an explicit commutator — Investigation #2 confirms. Composite v0.5 additive form retired (Reading (a) + better-fitting single object). T190’s own mechanism ($24 = N_c \cdot |W(B_2)|$ Weyl orbit) is the single-object substrate form.
- Tier: F50 v0.1 Investigation #1 closure; CKM Direction B FORWARD; muon $\rightarrow$ T190 (additive RETIRED) pending Investigation #2 explicit commutator.

7. Closure

Investigation #1 driven to substantive closure. The FK matrix element factors oppositely in the two sectors: the CKM mixing is irreducibly two-point (N_c^2 amplitude from two independent endpoint color traces, $= \lambda$ exactly; single-trace N_c^1 rejected), confirming Direction B; the muon mass operator is a single H^2 object (T190 $= (24/\pi^2)^{\{C_2\}}$, 0.003%), with the additive Composite v0.5 form retired under Reading (a) (coupled gradings + double-count + worse fit). This refines “one computation decides both” into two analogous-but-separate factorization questions with opposite answers in different spaces (color vs H^2 grading). The muon’s final confirmation is Investigation #2’s explicit [H_B, P_restriction] commutator (next, with Elie numerical); the CKM closure is firm. And the day-long muon thread resolves honestly: the additive form I built (F32/F45) gives way to the simpler single-object T190 it always sat near — the discipline retiring my own construction in favor of the cleaner one.

— Lyra, Investigation #1 FORWARD closure. CKM = TWO-POINT/Direction B (N_c^2 amplitude two independent endpoint color traces $= \lambda=9/40$ exactly; single-trace $N_c^1=3/40=0.075$ REJECTED). Muon = SINGLE-OBJECT T190 $(24/\pi^2)^{\{C_2\}}=206.761 @0.003\%$; Composite v0.5 additive 207@0.112% RETIRED under Reading (a) — 3 converging lines: (i) Reading (a) retires bulk+boundary additivity, (ii) [H_B,P_restriction] $\neq 0$ structurally (noncompact $SO(3,1) \not\supset K$ mixes compact K-types $\rightarrow$ gradings COUPLED not orthogonal $\rightarrow$ additive “+” illegitimate), (iii) T190 single-object fits 40 $\times$ better + Reading-(a)-safe ($24=N_c \cdot |W(B_2)|$ Weyl orbit, no additive split, colorless per F48). Refines Elie “one computation decides both” $\rightarrow$ two ANALOGOUS factorization questions, OPPOSITE answers, different spaces (CKM color two-point; muon H^2 -grading single-object). CKM/Wall 5 CLOSED Direction B ($\rightarrow$ Grace $R-6/d^2$ =second-moment branch, Keeper K234). Muon/Walls 2,3: T190 strongly indicated, additive RETIRED,

final confirm = Investigation #2 explicit [H_B,P_restriction] commutator (Lyra setup + Elie numerical on $V_{(1/2,1/2)}, V_{(3/2,1/2)}, V_{(5/2,1/2)}$; Keeper K232). Precision corroborates not forces (Cal #35; 0.003% below Two-Tier floor, weighted as corroboration).